Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

$\lim_{x \to 0} \left(\frac{x}{\tan^{-1} 2x} \right)$ is equal to

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) ∞

Q.2

$\lim_{x \to 0} \frac{x \tan 2x - 2x \tan x}{(1 - \cos 2x)^2}$ is

(A) 2

(B) -2

(C) $\frac{1}{2}$

(D) $-\frac{1}{2}$

Q.3

$\lim_{x \to \infty} (\sqrt{x + \sqrt{x + \sqrt{x}}} - \sqrt{x})$ is given by

(A) 0

(B) $\frac{1}{2}$

(C) $\log 2$

(D) none of these

Q.4

$\lim_{x \to 0} \left[\left(3x + \frac{1}{x}\right)^2 - \left(2x - \frac{1}{x}\right)^2 \right]$ is equal to

(A) 5

(B) 2

(C) 0

(D) 10

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Q.5

If $\lim_{x\to\infty} f(x)$ exists and is finite and $\lim_{x\to\infty}\left(f(x)+\frac{3f(x)-1}{(f(x))^2}\right)=3$, then the value of $\lim_{x\to\infty} f(x)$ is

(A) 1

(B) -1

(C) 2

(D) none of these

Q.6

$\lim_{x\to-\infty}\frac{x^3-x}{\sqrt{1+x^6}}$ is equal to

(A) -1

(B) 1

(C) 0

(D) 3

Q.7

$\lim_{x\to0}\left\{\frac{\log_e(1+x)}{x^2}+\frac{x-1}{x}\right\}=$

(A) ∞

(B) $\frac{1}{2}$

(C) $-\frac{1}{2}$

(D) none of these

Q.8

$\lim_{x\to0}\frac{\sin\left(\pi\cos^2 x\right)}{x^2}$ equals

(A) $-\pi$

(B) π

(C) $\pi/2$

(D) 1

Q.9

$\lim_{x\to2}\frac{(\cos\alpha)^x+(\sin\alpha)^x-1}{x-2}$ is equal to

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(A) ∞

(B) $(\cos\alpha)^2 + (\sin\alpha)^2$

(C) 0

(D) none of these

Q.10

The value of $\lim_{x\to 0} \frac{1}{x} \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ is

(A) 2

(B) ∞

(C) does not exist

(D) none of these

Q.11

$\lim_{x\to 0} \frac{\sqrt{\frac{1}{2}(1-\cos 2x)}}{x}$ is equal to

(A) 1

(B) -1

(C) 0

(D) none of these

Q.12

$\lim_{x\to\infty} \left(\frac{x^2+5x+3}{x^2+x+3}\right)^x$

(A) e^4

(B) e^2

(C) e^3

(D) e

Q.13

$\lim_{n\to\infty} \left\{\frac{1}{1-n^2} + \frac{2}{1-n^2} + \ldots + \frac{n}{1-n^2}\right\}$ is equal to

(A) 0

(B) $-1/2$

(C) $1/2$

(D) 1

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Q.14

$\lim_{n\to\infty}\left(3^n+5^n+7^n\right)^{1/n}$ is equal to

(A) e^3

(B) e^5

(C) 5

(D) 7

Q.15

Let $f(x)=\left(\frac{x}{x+3}\right)^{3x}$, then

(A) $\lim_{x\to\infty} f(x)=e^{-3}$

(B) $\lim_{x\to\infty} f(x)=e^{-9}$

(C) $\lim_{x\to 2} f(x)=\frac{64}{625}$

(D) none of these

Q.16

$\lim_{x\to 0}\frac{e^x+e^{-x}-2}{x^2}$ is equal to

(A) 1

(B) -1

(C) $\frac{1}{2}$

(D) $-\frac{1}{2}$

Q.17

$\lim_{x\to\infty}\frac{x^n}{e^x}=0$ is true, only when

(A) no value of n

(B) n is any whole number

(C) $n=0$ only

(D) $n=2$ only

Q.18

If $\lim_{x\to a}\frac{f(x)}{x}=l_1$ and $\lim_{x\to\frac{a}{\lambda}}\frac{f(\lambda x)}{x}=l_2$ then λ equal to

(A) 1

(B) $\frac{l_1}{l_2}$

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(C) $\frac{l_2}{l_1}$

(D) none of these

Q.19

The value of $\lim_{x\to 0}\left(\left[\frac{11x}{\sin x}\right]+\left[\frac{21\sin x}{x}\right]\right)$, where $[x]$ is the greatest integer less than or equal to x is

(A) 32

(B) 31

(C) 11

(D) 21

Q.20

$\lim_{x\to a-}\left(\frac{|x|^3}{a}-\left[\frac{x}{a}\right]^3\right)(a>0)$, where $[x]$ denotes the greatest integer less than or equal to x is

(A) a^2-3

(B) a^2-1

(C) a^2

(D) none of these

Q.21

If $f(x)$ is a polynomial satisfying $f(x)f\left(\frac{1}{x}\right)=f(x)+f\left(\frac{1}{x}\right)$ and $f(2)>1$, then $\lim_{x\to 1} f(x)$ is

Q.22

The value of $\lim_{x\to 1^-}\left[\sin^{-1}x\right]$ is (where [.] denotes greatest integer function)

Q.23

$\lim_{n\to\infty}\left\{\frac{2}{3}+\left(\frac{2}{3}\right)^2+\left(\frac{2}{3}\right)^3+\ldots\ldots\ldots+\left(\frac{2}{3}\right)^n\right\}$ is equal to

Q.24

If $\lim_{x\to\infty}\left(\sqrt{x^2-x+1}-ax-b\right)=0$ then for $k\geq 2, \lim_{n\to\infty}\sec^{2n}(k!\pi b)$ is equal to

Q.25

Let $f:R\to R$ be a positive increasing function with $\lim_{x\to\infty}\frac{f(3x)}{f(x)}=1$. Then $\lim_{x\to\infty}\frac{f(2x)}{f(x)}$

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Q.26

Find all possible value of a, if $\lim_{\mathrm{x}\to\mathrm{a}}\left(\frac{\mathrm{x}^9-\mathrm{a}^9}{\mathrm{x}-9}\right)^{\mathrm{x}}=\lim_{\mathrm{x}\to 3}(6+\mathrm{x})$

Q.27

The value of $\lim_{x\to 0}(\cos x)^{\cot\left(x^2\right)}$ is $\frac{n}{\sqrt{e}}$ then $n=$

Q.28

If $n>m$ are positive integers, then $\lim_{\alpha\to 0}\frac{\sin\alpha^n}{(\sin\alpha)^m}=$

Q.29

The least positive integer n for which the $\lim_{x\to 0}\frac{(\cos x-1)(\cos x-e^x)}{x^n}$ is a finite non-zero number is

Q.30

Let $f:\mathbf{R}\to[0,\infty)$ be such that $\lim_{x\to 5}f(x)$ exists and $\lim_{x\to 5}\frac{(f(x))^2-9}{\sqrt{|x-5|}}=0$ Then $\lim_{x\to 5}f(x)$ equals

Answers & Solutions

Q.1 Answer:

$\frac{1}{2}$

Solution:

Let $\tan^{-1}2x=\theta \quad\Rightarrow\quad x=\frac{1}{2}\tan\theta$ **and as** $x\to 0,\theta\to 0$

$\Rightarrow \quad \lim_{x\to 0}\frac{x}{\tan^{-1}2x}=\lim_{\theta\to 0}\frac{\frac{1}{2}\tan\theta}{\theta}=\frac{1}{2}$

Q.2 Answer:

$\frac{1}{2}$

Solution:

Required limit$=\lim_{x\to 0}\frac{x\tan 2x-2x\tan x}{(2\sin^2 x)^2}=\lim_{x\to 0}\frac{x\cdot\frac{2\tan x}{1-\tan^2 x}-2x\tan x}{4\sin^4 x}$

$=\lim_{x\to 0}\frac{2x\tan x}{(1-\tan^2 x)}\frac{\left(1-1+\tan^2 x\right)}{4\sin^4 x}=\lim_{x\to 0}\frac{2x\left(\frac{\tan x}{x}\right)^3\cdot x^3}{4(1-\tan^2 x)\left(\frac{\sin x}{x}\right)^4\cdot x^4}=\frac{1}{2}$

Q.3 Answer:

$\frac{1}{2}$

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Solution:

$\lim_{x\to\infty}(\sqrt{x+\sqrt{x+\sqrt{x}}}-\sqrt{x})=\lim_{x\to\infty}\frac{\sqrt{x+\sqrt{x}}}{\sqrt{x+\sqrt{x+\sqrt{x}}+\sqrt{x}}}=$

$\lim_{x\to\infty}\frac{\sqrt{1+x^{-1/2}}}{\sqrt{1+\sqrt{x^{-1}+x^{-3/2}}+1}}=\frac{1}{2}$

Q.4 Answer:

10

Solution:

$\lim_{x\to0}\left[\left(3x+\frac{1}{x}\right)^2-\left(2x-\frac{1}{x}\right)^2\right]$

$=\lim_{x\to0}\left[\left(3x+\frac{1}{x}-2x+\frac{1}{x}\right)\left(3x+\frac{1}{x}+2x-\frac{1}{x}\right)\right]=\lim_{x\to0}\left[\left(x+\frac{2}{x}\right).5x\right]=$
$\lim\left(5x^2+10\right)=10$

Q.5 Answer:

1

Solution:

Let $\lim_{x\to\infty}f(x)=l$

$\therefore\quad l+\frac{3l-1}{l^2}=3\quad\Rightarrow l^3-3l^2+3l-1=0\Rightarrow\quad(l-1)^3=0\Rightarrow l=1$

Q.6 Answer:

-1

Solution:

$\lim_{x\to-\infty}\frac{x^3-x}{\sqrt{1+x^6}}=\lim_{y\to\infty}\frac{-y^3+y}{\sqrt{1+y^6}}=\lim_{y\to\infty}\frac{y^3\left(-1+\frac{1}{y^2}\right)}{y^3\sqrt{\frac{1}{y^6}+1}}=\frac{-1}{1}=-1$

Q.7 Answer:

$\frac{1}{2}$

Solution:

Required limit$=\lim_{x\to0}\frac{\log_e(1+x)+x^2-x}{x^2}=\lim_{x\to0}\frac{\left(x-\frac{x^2}{2}+\frac{x^3}{3}-\ldots\ldots\right)+x^2-x}{x^2}$

$=\lim_{x\to0}\frac{\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\ldots\ldots}{x^2}=\lim_{x\to0}\frac{x^2\left(\frac{1}{2}+\frac{x}{3}-\ldots\ldots\right)}{x^2}=\frac{1}{2}$

Q.8 Answer:

π

Solution:

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$\lim_{x\to 0} \frac{\sin\left(\pi \cos^2 x\right)}{x^2} = \lim_{\to 0} \frac{\sin\left(\pi\left(1-\sin^2 x\right)\right)}{x^2}$

$= \lim_{x\to 0} \frac{\sin\left(\pi \sin^2 x\right)}{x^2} = \pi \lim_{x\to 0} \frac{\sin\left(\pi \sin^2 x\right)}{\pi \sin^2 x} \times \frac{\sin^2}{x^2}$

$= \pi \times 1 \times \mathbf{1} = \pi$

Hence, option 'B' is correct.

Q.9 Answer:

none of these

Solution:

$\lim_{x\to 2} \frac{(\cos\alpha)^x + (\sin\alpha)^x - 1}{x-2} = \lim_{x\to 2} \frac{(\cos\alpha)^x \log_e(\cos\alpha) + (\sin\alpha)^x \log_e(\sin\alpha)}{1}$

Using L'Hospital rule

$= \cos^2\alpha\left(\log_e \cos\alpha\right) + \sin^2\alpha \log_e(\sin\alpha)$

Q.10 Answer:

2

Solution:

Using $\sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\tan^{-1} x$**, the given limit equals.**

$\lim_{x\to 0} \frac{2\tan^{-1} x}{x} = 2\lim_{x\to 0} \frac{\tan^{-1} x}{x} = 2(1) = 2$

Q.11 Answer:

none of these

Solution:

$\lim = \lim_{x\to 0} \frac{|\sin x|}{x}$

R.H.L. $= \lim_{h\to 0} \frac{|\sin(0+h)|}{0+h} = \lim_{h\to 0} \frac{\sin h}{h} = 1$

L.H.L. $= \lim_{h\to 0} \frac{|\sin(0-h)|}{0-h} = \lim_{h\to 0} \frac{|-\sin h|}{-h} = -\lim_{h\to 0} \frac{\sin h}{h} = -1$

Since R.H.L. $\neq$ **L.H.L.**

$\therefore$ **limit does not exist.**

Q.12 Answer:

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e^4

Solution:

$\lim_{x\to\infty}\left(\frac{x^2+5x+3}{x^2+x+3}\right)^x = \lim_{x\to\infty}\left(1+\frac{4x}{x^2+x+3}\right)^x$

$= \lim_{x\to\infty}\left(1+\frac{4x}{x^2+x+3}\right)^{\frac{x^2+x+3}{4x}\times\frac{4x^2}{x^2+x+3}}$

$= \lim_{x\to\infty}\left(1+\frac{4x}{x^2+x+3}\right)^{\frac{x^2+x+3}{4x}\times\frac{4}{1+1/x+3/x^2}}$

$= e^4.$

Q.13 Answer:

$-1/2$

Solution:

$\frac{1}{1-n^2}(1+2+\ldots+n) = \frac{n(n+1)}{2(1-n^2)} = \frac{1}{2(1/n-1)}$

Q.14 Answer:

7

Solution:

$\lim_{n\to\infty}\left(3^n+5^n+7^n\right)^{1/n} \Rightarrow \lim_{n\to\infty} 7\left(\left(\frac{3}{7}\right)^n+\left(\frac{5}{7}\right)^n+1\right)^{1/n} = 7$

as $\left(\frac{3}{7}\right)^n$ and $\left(\frac{5}{7}\right)^n \to 0$ as n $\to \infty$

Q.15 Answer:

$\lim_{x\to\infty} f(x) = e^{-9}$

Solution:

$f(x) = \left(\frac{x}{x+3}\right)^{3x} = \frac{1}{\left(1+\frac{3}{x}\right)^{3x}}$

$\therefore \lim_{x\to\infty} f(x) = \frac{1}{\lim_{x\to\infty}\left[\left(1+\frac{3}{x}\right)^{x/3}\right]^9} = \frac{1}{e^9} = e^{-9}$

Q.16 Answer:

1

Solution:

$\lim_{x\to 0}\frac{e^x+e^{-x}-2}{x^2}$ $\left(\frac{0}{0}\right.$ **form)**

$\lim_{x\to 0}\frac{e^x-e^{-x}}{2x}$ $\left(\frac{0}{0}\right.$ **form)**

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$\lim_{x\to 0} \frac{e^x+e^{-x}}{2} = \frac{1+1}{2} = 1$

Q.17 Answer:

n is any whole number

Solution:

$\lim_{x\to\infty} \frac{x^n}{e^x} = \lim_{x\to\infty} n\frac{x^{n-1}}{e^x} = \ldots\ldots = \lim_{x\to\infty} \frac{n!}{e^x} = \frac{n!}{\infty} = 0$

where n is any whole number

($\because n!$ is defined only for all positive integers including zero)

Q.18 Answer:

$\frac{l_2}{l_1}$

Solution:

$\lim_{x\to a} \frac{f(x)}{x} = l_1, \lim_{x\to \frac{a}{\lambda}} \frac{f(\lambda x)}{x} = l_2$

$\Rightarrow \quad \lambda \lim_{t\to a} \frac{f(t)}{t} = \lambda l_1 = l_2 \quad \Rightarrow \quad \lambda = \frac{l_2}{l_1}$

Q.19 Answer:

31

Solution:

Since $x > \sin x$, for $x > 0$ and $\lim_{x\to 0+} \sin x = 1$

so $\frac{11x}{\sin x} \to 11$ as $x \to 0+$ but $\frac{11x}{\sin x} > 11$. Thus

$\left[\frac{11x}{\sin x}\right] = 11$ **for value $x \to 0+$ Similarly**

$\frac{21\sin x}{x} \to 21$ **as $x \to 0+$ but** $20 \le \frac{21\sin x}{x} < 21$

$\left[\frac{21\sin x}{x}\right] = 20$. **Hence**

$\lim_{x\to 0+} \left(\left[\frac{11x}{\sin x}\right] + \left[\frac{21\sin x}{x}\right]\right) = 31$
Similarly $x < \sin x$ for $x < 0$, so $\left[\frac{11x}{\sin x}\right] = 11$
as $x \to 0-$ and $\left[21\frac{\sin x}{x}\right] = 20$ as $x \to 0-$.
Thus $\lim_{x\to 0-} \left(\left[\frac{11x}{\sin x}\right] + \left[\frac{21\sin x}{x}\right]\right) = 31$

Q.20 Answer:

a^2

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Solution:

For, $0 < x < a$, $\left[\frac{x}{a}\right] = 0$.

So $\lim_{x \to a-} \left(\frac{|x|^3}{a} - \left[\frac{x}{a}\right]^3\right) = \frac{a^3}{a} = a^2$.

Q.21 Answer:

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Solution:

$f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right), f(2) > 1$

Let $f(x) = a_0 x^n + a_1 x^{n-1} + \ldots\ldots + a_{n-1} x + a_n, a \neq 0$

Then by comparing the coefficients of like powers, we get

$$a_n = 1, a_0^2 = 1, a_1 = a_2 = \ldots\ldots = a_{n-1} = 0$$

$\therefore \quad f(x) = -x^n + 1$ **or** $f(x) = x^n + 1$

Then $f(2) > 1 \Rightarrow f(x) = x^n + 1$

$\therefore \quad \lim_{x \to 1} f(x) = \lim_{x \to 1} (x^n + 1) = 2$

Q.22 Answer:

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Solution:

$\left\lfloor \frac{\pi}{2} \right\rfloor = 1$ **since LHL of** $\sin^{-1} x$ **is** $\frac{\pi}{2}$ **when** x **goes to 1.**

Q.23 Answer:

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Solution:

$\lim_{n \to \infty} \left\{\frac{2}{3} + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^3 + \ldots\ldots + \left(\frac{2}{3}\right)^n\right\}$

$= \lim_{n \to \infty} \frac{\frac{2}{3}\left(1 - \left(\frac{2}{3}\right)^n\right)}{1 - \frac{2}{3}} = \lim_{n \to \infty} 2\left\{1 - \left(\frac{2}{3}\right)^n\right\} = 2(1 - 0) = 2$ **(as** $n \to \infty, \left(\frac{2}{3}\right)^n \to 0$**)**

Q.24 Answer:

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Solution:

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$\lim_{x\to\infty}\left(\sqrt{x^2-x+1}-ax-b\right)=0$ **[Here** $a>0$ **for if** $a\le 0$**, then limit** $=\infty$]

$\lim_{x\to\infty}\frac{x^2-x+1-(ax+b)^2}{\sqrt{x^2-x+1}+ax+b}=0 \quad \lim_{x\to\infty}\frac{\left(1-a^2\right)x^2-(1+2ab)x+1-b^2}{\sqrt{x^2-x+1}+ax+b}=0$

This is possible only when $1-a^2=0$ **and** $1+2ab=0 \therefore \quad a=1(\mathrm{Q}a>0)$ **and** $b=-1/2$

Now $k!\pi b=k!\pi(-1/2)=$ **an integral multiple of** π **as** $k\ge 2$

$\therefore \quad \sec^2 k!\pi b=1; \quad \therefore \quad \lim_{n\to\infty}\left(\sec^{2n} k!\pi b\right)=1=a$

Q.25 Answer:

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Solution:

As f **is an increasing function, and**

$x<2x<3x\forall x>0$

we get

$f(x)<f(2x)<f(3x)\forall x>0$

$f(x)>0]$

$\Rightarrow 1<\frac{f(2x)}{f(x)}<\frac{f(3x)}{f(x)}$

Since $\lim_{x\to\infty}1=1$ **and**

$\lim_{x\to\infty}\frac{f(3x)}{f(x)}=1$ **(given), by the Sandwich theorem,**

we get $1\le\lim_{x\to\infty}\frac{f(2x)}{f(x)}\le 1\Rightarrow\lim_{x\to\infty}\frac{f(2x)}{f(x)}=1.$

Q.26 Answer:

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Solution:

Given: $\lim_{x\to a}\left(\frac{x^9-a^9}{x-9}\right)^x=\lim_{x\to 3}(6+x)$

To find: All possible values of a

Sol: $\lim_{x\to a}\left(\frac{x^9-a^9}{x-9}\right)^x=\lim_{x\to 3}(6+x)$

For limit to exist, it will have to have some uncertainity

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$\implies \lim_{x\to a}\left(\frac{x^9-a^9}{x-9}\right)=\frac{a^9-a^9}{a-9}$

Possible if $\mathrm{a}=9$

Hence $\mathrm{a}=9$

Q.27 Answer:

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Solution:

$\lim_{x\to 0}[1-(1-\cos x)]\left(\frac{1}{1-\cos x}\right)(1-\cos x)\cot\left(x^2\right)=e^{-\lim\frac{2\sin^2(x/2)\left(x^2/4\right)}{\left(x^2/4\right)\tan\left(x^2\right)}}=\frac{1}{\sqrt{e}}$

Q.28 Answer:

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Solution:

If $n>m$**, then** $\frac{\sin\alpha^n}{(\sin\alpha)^m}=\frac{\sin\alpha^n}{\alpha^n}\cdot\frac{\alpha^n}{\alpha^m}\left(\frac{\alpha}{\sin\alpha}\right)^m$

$=\frac{\sin\alpha^n}{\alpha^n}\cdot\alpha^{n-m}\left(\frac{\alpha}{\sin\alpha}\right)^m$

$\Rightarrow \lim_{\alpha\to 0}\frac{\sin\alpha^n}{(\sin\alpha)^m}=0$

Q.29 Answer:

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Solution:

Let $l=\lim_{n\to 0}\frac{(\cos x-1)(\cos x-e^x)}{x^n}=\lim_{n\to 0}\frac{-2\sin^2\frac{x}{2}(\cos x-e^x)}{x^n}$

For $n=1, n=2, l=0$

For $n=3$**,**

$l=\lim_{n\to 0}-2\left(\frac{\sin(x/2)}{x/2}\right)^2\cdot\frac{\cos x-e^x}{x^2\cdot x}=-\frac{1}{2}\lim_{n\to 0}\frac{\cos x-e^x}{x}$

$=\frac{1}{2}\lim_{n\to 0}(-\sin x-e^x)=-\frac{1}{2}\left(-e^0\right)=\frac{1}{2}$

$\therefore l$ **is finite and non-zero. Hence** $n=3$**.**

Q.30 Answer:

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Solution:

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As $\lim_{x\to 5} \frac{(f(x))^2-9}{\sqrt{|x-5|}} = 0$**, so**

$(f(x))^2 - 9 = K|x-5|^{\alpha}P(x)$**, where** $\alpha > 1/2$ **and** $P(x)$ **is a function of** x **such that** $\lim_{x\to 5} P(x)$ **exist** ϕ

$(\lim_{x\to 5}(f(x))^2 = \lim_{x\to 5}\left[K|x-5|^{\alpha}P(x) + 9\right] = 9$

The only possible answer is $\lim_{x\to 5} f(x) = 3$

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